

Experiment ‘p_1_3_product’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_1_3_product

Natural language statement: null

Method signature: p_1_3_product() returns (i: int)

Ensures

- `i == factorial(10)`

Functional Code Given

```
function factorial(n: int): int
  decreases n
{
  if n <= 1 then 1 else n * factorial(n - 1)
}
```

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_1_3_product()` returns (`i: int`)

The method should respect the following contract:

`ensures i == factorial(10)`

The contract uses the following dafny code:

```
function factorial(n: int): int
  decreases n
{
  if n <= 1 then 1 else n * factorial(n - 1)
}
```

These function(s) must not be used in your implementation of the method.

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn2CgWv91etbH7B3x06XE7AgmWNaA

Model: gpt-5.5-2026-04-23

Created at: 1780578914

Finish reason: stop

Usage: 348 tokens in, and 541 tokens out

```
{
  i := 3628800;
}
```

Final Program

```
function factorial(n: int): int
  decreases n
{
```

```
    if n <= 1 then 1 else n * factorial(n - 1)
  }

method p_1_3_product() returns (i: int)
  ensures i == factorial(10)
{
  i := 3628800;
}
```

Total Token Usage

Input tokens: 348

Output tokens: 541

Reasoning tokens: 512

Sum of 'total tokens': 889

Experiment Timings

Overall Experiment started at 1780578910348, ended at 1780578967792, lasting 57444ms (57.44 seconds)

Iteration #1 started at 1780578910521, ended at 1780578967791, lasting 57270ms (57.27 seconds)